Exercise - REST Connector

In this exercise, you will create a new *Entertain Me* process which contains a single *Generate Joke* task.

The *Generate Joke* task will use the **REST Connector** to obtain a joke from a publicly available Joke API.

Create Process

Create a new **Process Definition** by following the steps below:

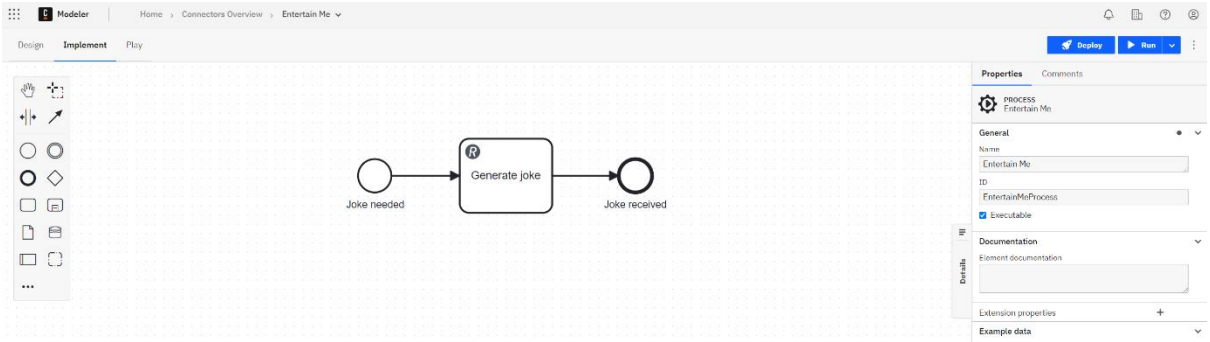
1. Launch the **Web Modeler**
2. Create a new **BPMN Diagram**
3. This will open a new **BPMN Diagram** in the **Modeler**. In the properties panel on the right-hand side of the screen, set the **Name** and **ID** as follows:

Name	Value
Name	Entertain Me
ID	EntertainMeProcess

   !!! info "Implement" Mode The Web Modeler offers different modes for easier collaboration between business and IT. To access the properties panel and later to deploy and start your process, make sure to stay in **Implement** mode for this exercise. Read more about the different modes here
4. Model a **Start Event**, **Task** and **End Event** with the following properties:

Element	Name	ID
Start Event	Joke needed	StartEvent_JokeNeeded
Task	Generate joke	Task_GenerateJoke
End Event	Joke received	EndEvent_JokeReceived

5. Your process should look like the diagram below once you have connected the sequence flows:



Configure the Generate Joke Task

You will now need to configure the *Generate Joke* Task to use the **REST Connector**.

1. Select the *Generate Joke* task, click on the **+Select** button in the **Template** section of the properties panel and choose **REST Connector** from the menu that appears.
2. Configure the **Authentication** properties as follows:

Name	Value
Type	None

Authentication

In this example, we are using an open API which does not require authentication.

3. Configure the **HTTP Endpoint** properties as follows:

Name	Value
Method	GET
URL	https://official-joke-api.appspot.com/jokes/programming/random

API Endpoint

In this example, we will send a GET request to the **Official Joke API**. We will use the `programming` category to request a `random` joke.

4. Configure the **Connection** properties as follows:

Name	Value

Connection Timeout	20

Connection Timeout

The default value for the request timeout is 20 seconds; an incident will be created in **Operate** if the request takes longer than this.

5. The snippet below contains an example response from a request to the Joke API. We will extract the relevant elements from this response in the next step.


```
{
  "status":200,
  "headers":
  {
    "date":"Thu, 01 Jan 1970 00:00:00 GMT",
    "access-control-allow-origin":"*",
    "server":"Google Frontend",
    "transfer-encoding":"chunked",
    "vary":"Accept-Encoding",
    "x-cloud-trace-context":"qwertzui1234567;o=1",
    "x-powered-by":"Express",
    "content-type":"application/json; charset=utf-8",
    "etag":"W/\"a1-qwert123456\"",
    "cache-control":"private",
    "alt-svc":"h3=\":443\"; ma=2592000,h3-29=\":443\"; ma=2592000,h3-
Q050=\":443\";"
  },
  "body":
  [
    {
      "type":"programming",
```

```
      "setup":"Why do C# and Java developers keep breaking their
keyboards?",

      "punchline":"Because they use a strongly typed language.",

      "id":74

    }

  ]

}
```

6. Configure the **Result Expression** of the **Response Mapping** property with the following JSON:

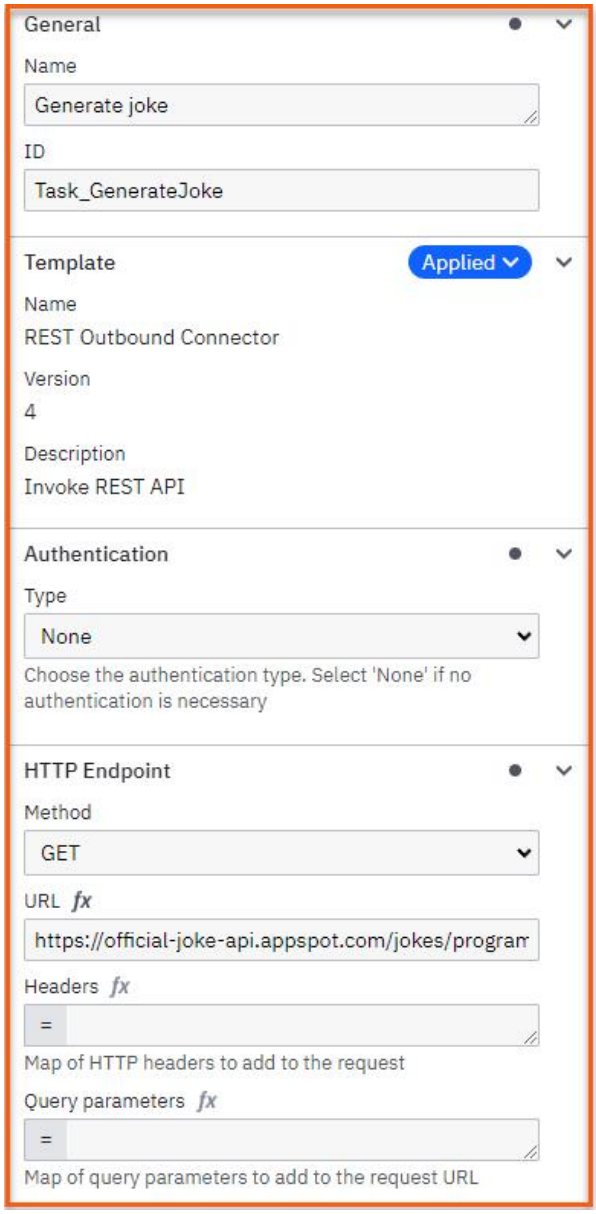
```
{

  "setup" : body.setup[1],

  "the punchline" : body.punchline[1]

}
```

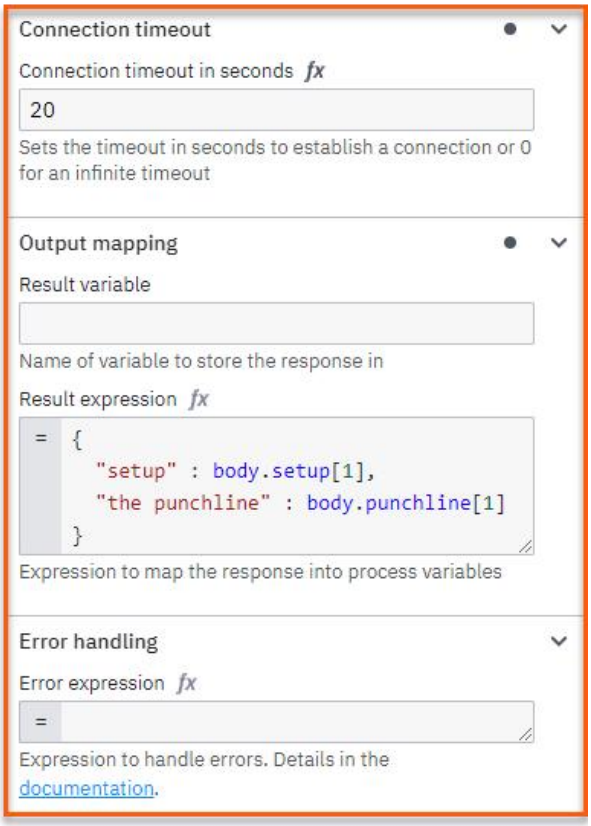
Result Expression

If we use a **Result Variable**, the Connector will simply store the entire response for the request in that variable. It is usually better to extract only the parts of the result which are relevant for the process. This is possible using the **Result Expression**.

In the **Result Expression**, we have defined two new variables (`setup` and `punchline`). The value of these variables is initialized from the first element of the JSON array returned in the body of the response.

7. The properties of the **Create Joke** task should look like this now:





Test the Process

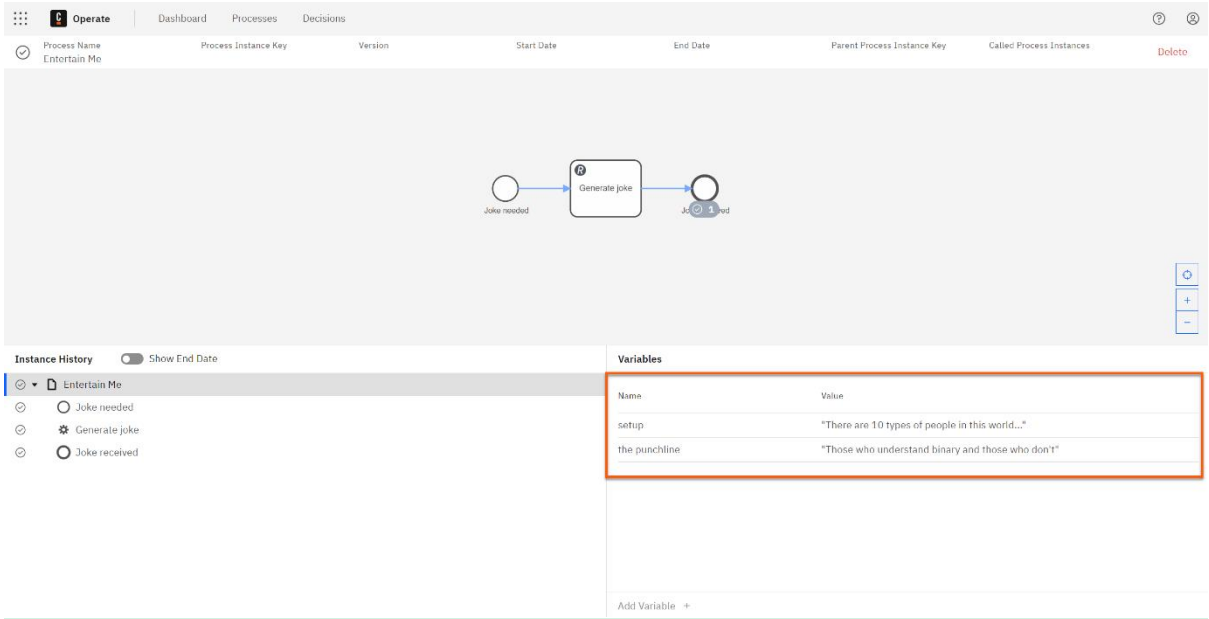
You can now test the **REST Connector** by deploying your process and starting an instance.

1. Deploy the process to your cluster by clicking the **Deploy Diagram** button
2. Click the **Run** button to start a new instance of the process
3. Launch **Operate** and select the *Entertain Me* process from **Process Instances by Name**

   Operate

   You can switch to **Operate** using the **App Switcher** in the top left corner of the screen. Alternatively, you can click on the **Process Started** pop-up at the bottom of the screen which is displayed for a few seconds after starting your instance.

4. Click on the **Process Instance Key** for the process that you have just started; it should have completed immediately
5. You should be able to see that your process instance has completed successfully and the `setup` and `punchline` variables have been populated as **Process Variables**.



Configured the REST Connector

Congratulations! You have created a process which uses the **REST Connector** to obtain data from an external service.

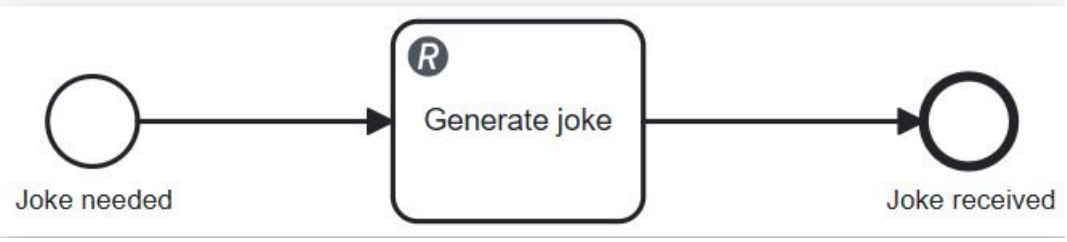
Solution - REST Connector

If you encountered any problems during this course or were unable to complete the process configuration, you can compare your solution with our example solution below.

GitHub Repository

Our version of the **Process Definition** including the **REST Connector** configuration can be found in the GitHub Repository below:

- Solution - Entertain Me

Process Diagram

Our version of the **Process Diagram** can be found below: